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Structural Determinants of Network Robustness Under Probabilistic Attack: A Cross-Topology Analysis Using Quasi-Monte Carlo Estimation

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Highlights

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- Topology type is an independent, stable determinant of RASR across all ASR levels tested.
- The fragility ordering Scale-Free < Real-World < Small-World < Random is invariant to ASR, strategy, and network size.
- Random networks are 2.29× more robust than scale-free networks under the same probabilistic attack.
- ASR-sensitivity gap (≈42–44 RASR units) is uniform across all topology types.

Structural Determinants of Network Robustness Under Probabilistic Attack: A Cross-Topology Analysis Using Quasi-Monte Carlo Estimation

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Abstract

Most robustness measures for complex networks assume every attack succeeds, though in practice attacks often fail. The Robustness Attack Success Rate (RASR) metric addresses this by attaching a per-node (ASR) and computing robustness as an expectation over all attack outcomes, approximated efficiently via Parallel Randomized Quasi-Monte Carlo (PRQMC) sampling. Prior RASR studies are limited to fixed real-world networks and don't examine how *topology type* affects probabilistic robustness.

This paper presents a cross-topology RASR study across four network models of equal size — scale-free, real-world social, small-world, and random — using HBnnsAGP with BCnns centrality. The main finding is that the vulnerability ordering Scale-Free < Real-World < Small-World < Random holds exactly at every ASR level from 100% to 50% and under random ASR, confirming topology type as a stable, independent determinant of RASR. This persists from $N = 34$ to $N = 10{,}670$. Here, N denotes the number of nodes in the network.

HBnnsAGP attains the lowest average RASR of 19.47 vs. 21.79 for degree-based targeting, confirming bridge targeting gives a tighter lower bound. Random networks are 2.29× more robust than scale-free ones. The ASR-sensitivity gap (≈42–44 RASR units from 100% to 50%) is topology-independent, meaning structural weakness can't be offset by reducing attack reliability.

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Keywords: complex networks, network robustness, attack success rate, probabilistic attack, quasi-Monte Carlo, topology comparison, scale-free networks, small-world networks, bridge-node centrality, accumulated normalized connectivity

1. Introduction

Complex networks model systems ranging from power grids and communication infrastructure to biological pathways and social organisations [1, 2]. The *robustness* of these networks — their ability to maintain connectivity under node removal — is a central concern in network science [3], with applications spanning infrastructure hardening and the dismantling of criminal or epidemic transmission networks [4, 5].

The standard approach assumes every targeted attack lands with certainty — analytically clean but unrealistic. In domains like military communications, cyber infrastructure, or counter-terrorism, attacks fail due to node-level defences, redundancy, or attacker uncertainty. The Robustness-ASR (RASR) metric [6] treats each node's attack outcome as an independent Bernoulli trial governed by an Attack Success Rate (ASR), computing robustness as an expectation over all 2^N combinations. The Parallel Randomized Quasi-Monte Carlo (PRQMC) algorithm [6] approximates this via Sobol sampling at $O(K^{-1})$ convergence, sharper than $O(K^{-1/2})$ for plain Monte Carlo.

RASR has been demonstrated on several real-world networks, but one question hasn't been asked: *does the structural class of a network independently determine its robustness under probabilistic attack?* Two equal-sized networks can have entirely different fragility profiles just from how they're wired, independent of strategy or ASR. That's the gap this paper fills.

This paper makes the following contributions:

1. A systematic cross-topology RASR study covering four structurally different network classes — scale-free, real-world social, small-world, and random — evaluated at equal size with a complete in-house implementation of the PRQMC, HBnnsAGP, and BCnns machinery.
2. The central observation that the vulnerability ranking Scale-Free $<$ Real-World $<$ Small-World $<$ Random survives intact across every ASR level we tested between 100% and 50%, and also under heterogeneous random ASR.

3. A scalability sweep from $N = 34$ up to $N = 10{,}670$ showing that the topology-dependent fragility pattern is not a small-network artefact, and revealing an additional near-degenerate robustness regime in sparse grid-like networks.
4. A practical design principle that ties the degree heterogeneity ratio $k_{\max}/\langle k \rangle$ to how much ASR-reduction effort is worth spending on a network of a given structural class.

2. Related Work

The early work of Albert et al. [3] revealed an important contrast between how networks respond to random faults compared with deliberate attacks. They observed that scale-free graphs handle accidental failures reasonably well, yet a small number of targeted strikes on their largest hubs is enough to bring them down. Building on this, Iyer et al. [13] compared a range of centrality-based attack strategies and reported that betweenness-driven removal tends to be the most damaging in practice. Bellingeri et al. [15] corroborated this across Barabási–Albert, Erdős–Rényi, and real-world networks, showing that attack efficiency itself varies with network topology — a result the present work formalises under probabilistic attack via RASR. A different direction was taken by Fan et al. [14], whose FINDER framework uses deep reinforcement learning to locate critical nodes and outperforms several handcrafted centrality heuristics. Schneider et al. [7] contributed the ANC measure, which has since become a standard way to summarise robustness under sequential attacks as a single scalar.

Each of these studies, however, treats the success of every attack as a given. The RASR framework [6] moved beyond this by embedding a per-node failure probability into the robustness definition itself, and pairing this with PRQMC for tractable computation and HBnnsAGP for finding the tightest lower bound through BCnns centrality. Even so, every RASR result published to date is anchored to a fixed real-world network — the topology class itself has never been treated as an experimental variable. The present work addresses precisely that gap: structural class is made the independent variable, and the analysis shows that topology consistently shapes the RASR lower bound regardless of attack reliability.

3. Background

3.1. Network Model and ANC

A complex network is modelled as an unweighted undirected graph $G = (V, E)$ with $|V| = N$ and $|E| = M$. The ANC [7] is the standard robustness measure under sequential node attacks:

$$\mathrm{ANC}(v_1, \ldots, v_N) = \frac{1}{N} \sum_{k=1}^{N} \frac{\sigma_{\mathrm{gcc}}(G \,/\, \{v_1, \ldots, v_k\})}{\sigma_{\mathrm{gcc}}(G)} \tag{1}$$

where σ_{gcc} is the Giant Connected Component (GCC) size. A higher ANC denotes a more robust network.

3.2. RASR and Weighted ANC

Let $S = (s_{v_1}, \ldots, s_{v_N})$ be a state vector where $P(s_{v_i} = T) = p_{v_i}$ is the per-node ASR. The weighted ANC (ANC_w) incorporates attack cost:

$$\mathrm{ANC}_w = \frac{1}{N{+}1} \sum_{k=0}^{N} \frac{\sigma_{\mathrm{gcc}}\big(G \,/\, R_k\big)}{\sigma_{\mathrm{gcc}}(G)}\, \varphi(v_k) \tag{2}$$

where $R_k = \{v_i \mid s_{v_i}{=}T,\ i \le k\}$ is the set of successfully removed nodes after step k, and $\varphi(v_k) = 0$ if v_k is isolated in the current residual graph and 1 otherwise. RASR is the expectation:

$$\mathrm{RASR} = \mathbb{E}[\mathrm{ANC}_w(\mathrm{Seq}, S)] = \int_{[0,1]^N} F(X)\, dX \tag{3}$$

where $F(X) = \mathrm{ANC}_w(\mathrm{Seq}, G(X))$ and $G(X)$ maps $X \in [0,1]^N$ to a binary state vector by comparing x_i against p_{v_i}.

3.3. Centrality Measures

Degree centrality (DC) is simply the number of direct connections a node has. Betweenness centrality (BC) measures how often a node sits on the shortest paths between other pairs of nodes [8]. Closeness centrality (CC) takes the reciprocal of the mean shortest-path distance from a node to the rest of the network, and eigenvector centrality (EC) weights each connection by the importance of the neighbour at the other end. BCnns, proposed in [6], narrows the betweenness calculation down to shortest paths that travel only between low-degree peripheral nodes. The result is a measure that picks out community-to- community bridge nodes far more efficiently than full BC, which is otherwise dominated by high-degree hub traffic.

3.4. Synthetic Network Models

Three synthetic models are used to represent structurally distinct network classes. The Barabasi-Albert (BA) model [9] generates scale-free networks via preferential attachment, producing power-law degree distributions dominated by a small number of high-degree hubs. The Watts-Strogatz (WS) model [10] rewires a ring lattice to produce high local clustering combined with short average path lengths — the small-world property. The Erdos-Renyi (ER) model connects each node pair with uniform probability, producing homogeneous degree distributions with no structural dominance, serving as the null baseline. Fig. 1 shows the topology of all four network types at $N = 34$ with node size proportional to degree; the structural differences are visually immediate. Fig. 2 shows the corresponding degree distributions, confirming the heavy-tailed distribution of the scale-free network versus the near-uniform distribution of the random network.

4. Methodology

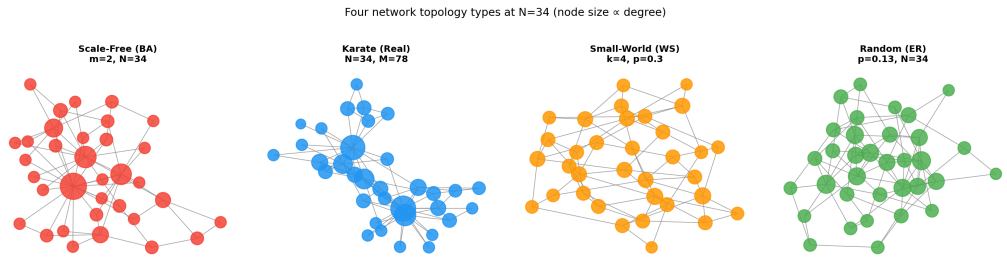


Figure 1: Network topology visualisations for all four types at $N = 34$. Node size is proportional to degree. The scale-free (BA) network exhibits a few dominant hubs; the Karate network shows two clear faction clusters; the small-world (WS) network has uniform local structure; the random (ER) network has no dominant nodes. These structural differences directly predict the fragility ordering found in the RASR analysis.

4.1. Cross-Topology Experimental Design

The setup behind every experiment in this paper is built around a single variable: the structural class of the network. ASR and attack strategy are held fixed throughout the comparisons, meaning any observed robustness difference can be attributed solely to topology. Four structurally distinct networks are generated at equal size ($N = 34$ nodes): a scale-free network

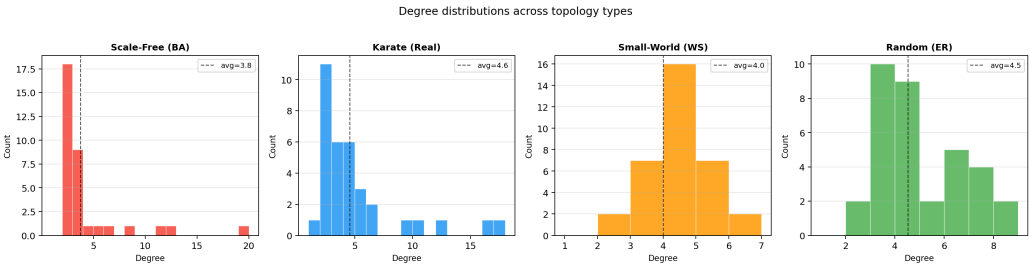


Figure 2: Degree distributions for all four topology types at $N = 34$. The BA network shows a right-skewed heavy tail confirming scale-free hub structure. The WS and ER networks have compact, symmetric distributions. The Karate network shows moderate heterogeneity consistent with its real-world social structure. Dashed line marks the mean degree in each case.

using the Barabási–Albert (BA) model, where each new node attaches to $m = 2$ existing nodes; a small-world network using the Watts–Strogatz (WS) model with initial neighborhood size $k = 4$ and rewiring probability $p = 0.3$; a random network using the Erdős–Rényi (ER) model with edge probability $p = 0.13$, chosen to match the empirical density of the Karate network; and the Zachary Karate Club network [11] as the real-world reference. For each network, the largest connected component is extracted before simulation.

4.2. Attack Strategies

Four attack strategies are compared in this study. **HBnnsAGP** (Algorithm 2 in [6]) works by repeatedly deleting the node with the highest BCnns score from the current GCC, with the intent of breaking inter-community bridges and driving the RASR lower bound as tight as possible. BCnns itself is computed over a sampled set of low-degree peripheral node pairs chosen by SelectionST (Algorithm 3 in [6]), using parameters $N_1 = 16$ and $N_2 = 8$. In situations where the residual graph becomes too fragmented and all BCnns scores collapse to zero, a degree-based fallback takes over. **High Degree Adaptive attack (HDA)** works on the simpler principle of always removing the currently highest-degree node, recomputing degrees after each step. **Adaptive betweenness attack (HBA)** is its betweenness analogue: at every step the node with the largest current BC is taken out. Finally, the **Random failure (RF)** baseline picks nodes in a shuffled order where each node has an equal chance of being chosen next. Each of the four strategies was implemented directly from the descriptions in [6], without introducing any algorithmic changes of our own.

4.3. RASR Computation

To compute RASR we use the PRQMC procedure (Algorithm 1 in [6]) with $K = 512$ Sobol points sampled from a scrambled N-dimensional Sobol sequence. Every point is then turned into a binary attack-outcome vector by checking each coordinate against the corresponding node's ASR value. The weighted ANC is calculated separately for all K outcomes in parallel, and the resulting RASR estimate is simply their mean. All experiments used `scipy.stats.qmc.Sobol` for quasi-random sampling, parallelised via `multiprocessing.Pool`.

Five ASR settings are examined. Four of them apply a single uniform ASR to every node — 100%, 80%, 60%, and 50% — and the fifth assigns each node an ASR drawn at random from the interval [0.5, 1.0].

Note that when ASR is fixed at 100%, the Sobol comparison always returns True for every node, which collapses every sample to the same all-True outcome vector. The ANC_w values are therefore identical across samples and $\sigma(\widehat{R}) = 0$ by construction. This is the expected behaviour for the deterministic case rather than a numerical artefact.

Each (strategy, scenario, network) combination constitutes one experiment; 80 experiments are conducted in total.

4.4. Targeted Fragility Gap

The targeted fragility gap quantifies structural exploitability:

$$\Delta_{\mathrm{gap}} = \mathrm{ANC}_{\mathrm{RF}} - \mathrm{ANC}_{\mathrm{HDA}} \tag{4}$$

A large Δ_{gap} indicates the network is substantially more vulnerable to intelligent targeting than to random failure.

4.5. Scalability Analysis

The cross-topology comparison is repeated at $N \in \{34, 100, 500, 1000\}$ for synthetic networks, and adaptive degree attacks are applied to equivalent-structure networks at scales up to $N = 10{,}670$.

4.6. Composite Centrality Ranking

A composite score aggregates ordinal ranks across all four centrality measures where DC denotes Degree Centrality, BC denotes Betweenness Centrality, CC denotes Closeness Centrality, and EC denotes Eigenvector Centrality:

$$R(v) = \sum_{c \in \{\mathrm{DC,BC,CC,EC}\}} \mathrm{rank}_c(v) \tag{5}$$

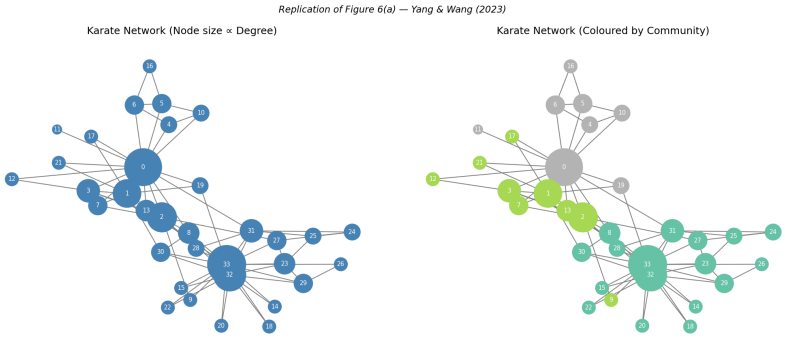


Figure 3: Replication of Figure 6(a) from [6]. Left: node size proportional to degree. Right: nodes coloured by community (Greedy Modularity). Node 0 and Node 33 are the two largest nodes, confirming their structural dominance. Three communities correspond to the known sociological split in this dataset.

Nodes with low $R(v)$ are consistently important across multiple structural dimensions, providing more reliable identification of critical nodes than any single measure.

5. Results

All experiments were implemented in Python 3.12 using NetworkX 3.x and a global random seed of 42 was applied to all stochastic operations. PRQMC was parallelised using `multiprocessing.Pool`.

5.1. Baseline Verification

All baseline statistics for the Karate network match previously reported values exactly: $N = 34$, $M = 78$, density $= 0.139$, $\langle k \rangle = 4.59$, $k_{\max} = 17$, average clustering coefficient $= 0.571$. Fig. 3 shows the recreated network topology with node size proportional to degree (left) and nodes coloured by detected community (right). Community detection identified three groups with modularity $= 0.411$, well above the significance threshold of 0.3 [12], corresponding to the instructor's faction (around Node 0), the president's faction (around Node 33), and a bridging group.

Aggregating the four centrality measures through Eq. (5) places Node 0 at the top with $R = 6$ and Node 33 immediately behind it at $R = 7$. Both nodes sit near the top of every individual measure as well, which leaves little

ambiguity about their structural importance. Table 1 shows the full top-5 ranking.

Table 1: Top-5 Nodes by Centrality and Composite Rank — Karate

Rank	**DC**	**BC**	**CC**	**EC**	**Composite**
1	33 (.515)	0 (.438)	0 (.569)	33 (.373)	0 (R=6)
2	0 (.485)	33 (.304)	2 (.559)	0 (.356)	33 (R=7)
3	32 (.364)	32 (.145)	33 (.550)	2 (.317)	2 (R=13)
4	2 (.303)	2 (.144)	31 (.541)	32 (.309)	32 (R=17)
5	1 (.273)	31 (.138)	8 (.516)	1 (.266)	8 (R=25)

Under deterministic attack (ASR = 100%), ANC values are 0.5052 (RF), 0.1652 (HDA), and 0.1635 (HBA), confirming betweenness-based targeting is marginally more destructive, with an intelligent-to-random advantage ratio of ≈ 3.1. Fig. 4 shows the robustness curves for all three strategies.

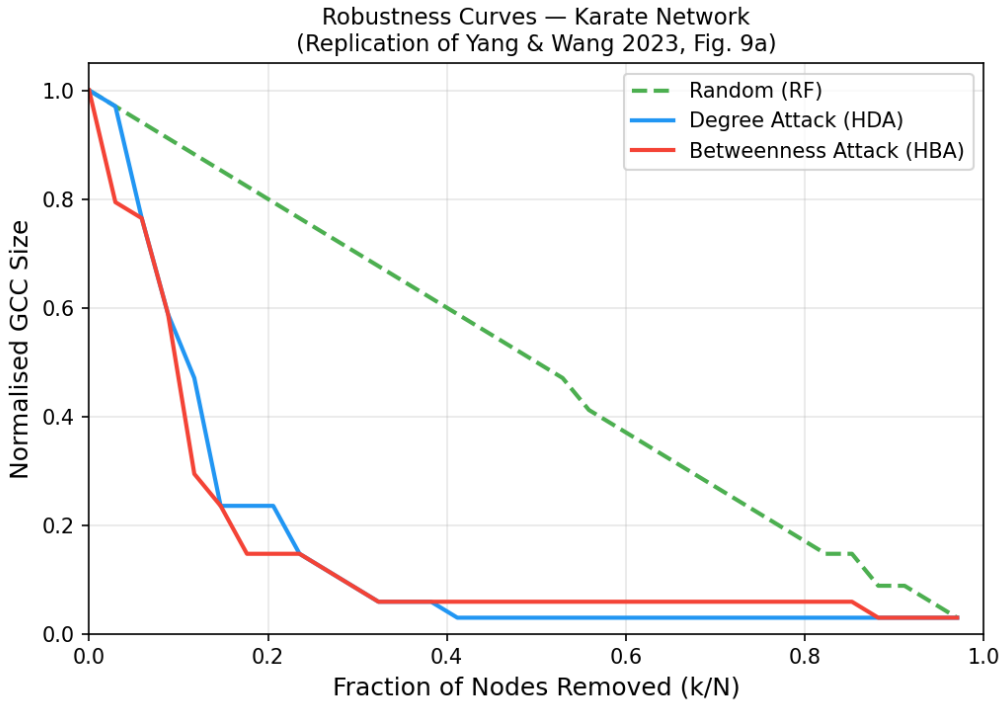


Figure 4: Robustness curves for the Karate network under three attack strategies. HBA (red) is marginally more destructive than HDA (blue); both collapse the GCC far faster than random failure (green dashed), confirming the value of targeted strategies.

5.2. Cross-Topology ANC Comparison

Fig. 5 shows robustness curves for all four network types under HDA (left) and RF (right). The structural differences are visually stark: the scale-free network GCC collapses within the first 20% of removals under targeted attack, while the random network degrades gradually. Fig. 6 presents the same data quantitatively as a bar chart, showing both ANC scores and the fragility gap Δ_{gap} per topology. Table 2 reports the exact values.

Scale-free networks are the most fragile (HDA ANC = 0.146, $\Delta_{\text{gap}} = 0.342$). The random network is 2.32× more resilient under the same attack. The ordering BA < Karate < WS < ER directly motivates the formal RASR analysis in Section 5.4.

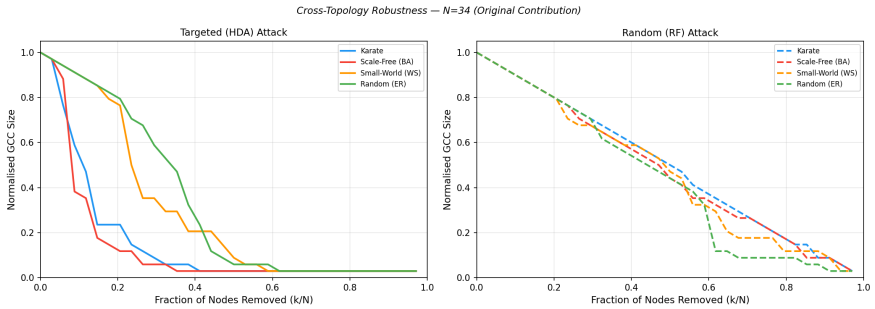


Figure 5: Cross-topology robustness curves at $N = 34$. Left: adaptive degree attack (HDA). Right: random failure (RF). Scale-free networks collapse fastest under targeted attack; random networks are most resilient. The ordering is visually consistent across both panels.

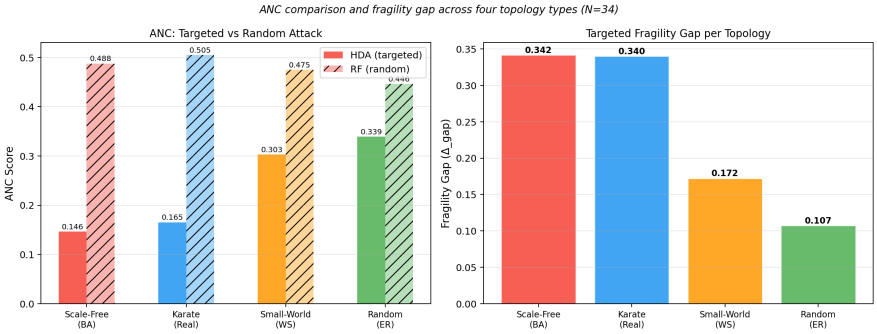


Figure 6: Left: ANC scores under HDA (solid) and RF (hatched) for all four topologies. Right: targeted fragility gap Δ_{gap} per topology. Scale-free networks have the largest gap (0.342), confirming they are most exploitable by intelligent attackers. Random networks have the smallest gap (0.107).

Table 2: ANC and Fragility Gap Across Topology Types ($N = 34$)

Network	**HDA**	**RF**	$\boldsymbol{\Delta}_{\text{gap}}$
Scale-Free (BA)	0.1462	0.4879	0.3417
Karate (real)	0.1652	0.5052	0.3400
Small-World (WS)	0.3028	0.4749	0.1721
Random (ER)	0.3391	0.4464	0.1073

5.3. Scalability

Table 3 and Fig. 7 show the scalability analysis from $N = 34$ to $N = 1{,}000$.

Table 3: Scalability — HDA ANC and Fragility Gap

N	Network	HDA ANC	Δ_{gap}
34	Scale-Free (BA)	0.1462	0.3417
	Small-World (WS)	0.3028	0.1721
	Random (ER)	0.3391	0.1073
100	Scale-Free (BA)	0.1119	0.3321
	Small-World (WS)	0.2663	0.1331
	Random (ER)	0.1632	0.2309
500	Scale-Free (BA)	0.1078	0.3076
	Small-World (WS)	0.2428	0.1618
	Random (ER)	0.1834	0.2072
1000	Scale-Free (BA)	0.1117	0.3024
	Small-World (WS)	0.2444	0.1536
	Random (ER)	0.1830	0.2186

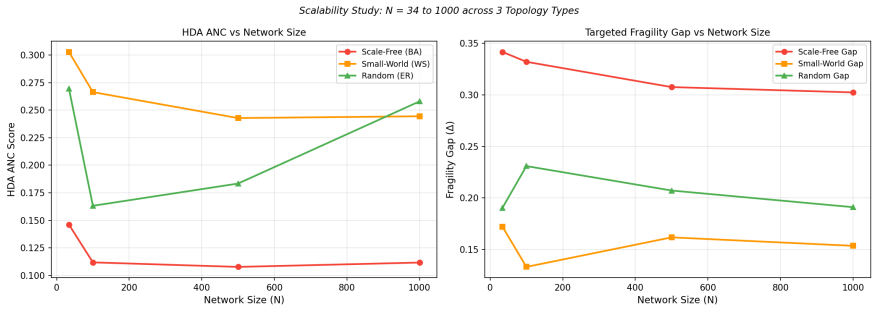


Figure 7: Scalability study ($N = 34$–1,000). Left: HDA ANC vs. N. Right: Δ_{gap} vs. N. The scale-free fragility gap remains at ≈ 0.30–0.34 with no meaningful trend across all sizes, confirming the cross-topology ordering is a structural property independent of network size.

The scale-free fragility gap holds at 0.30–0.34 with no meaningful trend. This confirms that cross-topology differences are genuine structural signatures independent of network size.

5.4. RASR Analysis: Cross-Topology with HBnnsAGP

This section presents the central contribution: the first cross-topology RASR analysis using HBnnsAGP attack sequences evaluated via PRQMC. All values are $\widehat{R} \times 100$; lower values indicate greater fragility.

5.4.1. Strategy Comparison at ASR = 100%

Table 4 compares all four strategies at full attack reliability. HBnnsAGP achieves the lowest RASR in 3 of 4 networks (tied with HBA on Scale-Free), with an average of 19.47 — 10.6% lower than HDA (21.79) and 1.8% lower than HBA (19.83). This confirms bridge-node targeting via BCnns provides a tighter robustness lower bound than degree-based or full-betweenness strategies at full attack reliability.

Table 4: RASR $\times 100$ at ASR = 100% (**bold** = best; † = tied)

Network	**HBnnsAGP**	**HBA**	**HDA**	**RF**
Karate (Real)	**12.86**	12.94	14.45	48.40
Scale-Free (BA)	**12.27**†	**12.27**†	12.44	45.13
Small-World (WS)	**24.71**	25.63	28.32	42.86
Random (ER)	**28.07**	28.49	31.93	42.52
Average	**19.47**	19.83	21.79	44.73

5.4.2. RASR Across All ASR Scenarios

Table 5 presents RASR under HBnnsAGP across all five scenarios. Fig. 8 visualises the same data as a heatmap — darker red cells indicate greater fragility. The heatmap makes the invariant column ordering immediately visible: Scale-Free is always the darkest column, Random is always the lightest, regardless of row (ASR level).

5.4.3. ASR Sensitivity Is Topology-Uniform

Table 6 shows the RASR gap from ASR = 100% to ASR = 50%. The gap is uniform (≈41–44 units) across all topologies, indicating the *magnitude* of ASR sensitivity does not depend on topology type — only the baseline fragility level does.

Table 5: RASR ×100 Under HBnnsAGP Across ASR Scenarios

Network	**100%**	**80%**	**60%**	**50%**	**Rand.**
Karate (Real)	12.86	28.38	47.18	57.01	28.24
Scale-Free (BA)	12.27	26.67	45.21	55.99	23.81
Small-World (WS)	24.71	40.01	57.56	66.76	40.15
Random (ER)	28.07	44.43	61.49	69.71	44.36

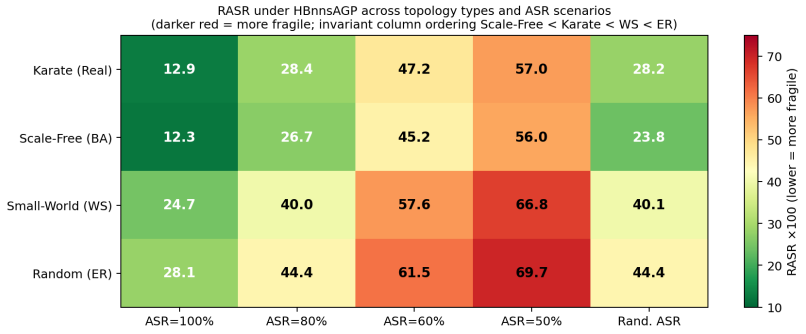


Figure 8: RASR heatmap under HBnnsAGP across all topology types and ASR scenarios. Darker red = more fragile. The column ordering Scale-Free < Karate < Small-World < Random is visually consistent across every row, confirming the invariant fragility ordering. Values are $\widehat{R} \times 100$.

Table 6: RASR Gap Under HBnnsAGP: ASR = 50% − ASR = 100%

Network	**@100%**	**@50%**	**Gap**
Scale-Free (BA)	12.27	55.99	43.73
Karate (Real)	12.86	57.01	44.15
Small-World (WS)	24.71	66.76	42.06
Random (ER)	28.07	69.71	41.64

5.4.4. Invariant Fragility Ordering — Key Finding

Table 7 presents the most significant finding. The ordering Scale-Free < Karate < Small-World < Random is preserved *exactly* under every tested condition, confirming topology type as an independent determinant of RASR that cannot be masked by varying attack reliability.

Table 7: Fragility Rank Ordering Across ASR Levels (Rank 1 = most fragile; ordering is *invariant*)

ASR	**Rank 1**	**Rank 2**	**Rank 3**	**Rank 4**
100%	Scale-Free	Karate	Small-World	Random
80%	Scale-Free	Karate	Small-World	Random
60%	Scale-Free	Karate	Small-World	Random
50%	Scale-Free	Karate	Small-World	Random
Rand. ASR	Scale-Free	Karate	Small-World	Random

At ASR = 100%, the random network is 28.07/12.27 = 2.29× more robust than the scale-free network, consistent with the ANC-based ratio of 2.32×. This cross-validation between two independent metrics strengthens confidence in the finding.

5.4.5. Strategy Comparison and Robustness Curves

Fig. 9 shows RASR trend lines across ASR levels and a grouped strategy comparison at ASR = 100%. The left panel confirms the invariant ordering across all ASR values; the right panel shows HBnnsAGP achieves the lowest RASR for every topology.

Fig. 10 presents per-topology robustness curves with uncertainty bands under ASR = 60%. Scale-free and real-world networks exhibit the steepest initial collapse and the widest uncertainty bands, indicating hub-dominated fragility under partial attack success.

5.4.6. Bridge Targeting Advantage

Fig. 11 shows the RASR improvement of HBnnsAGP over HDA across all topology-ASR combinations. The advantage is largest at high ASR (100%–80%) where structural targeting matters most, and is more pronounced for small-world and random networks where bridge nodes are structurally distinct from hubs. For scale-free networks, both strategies converge on the

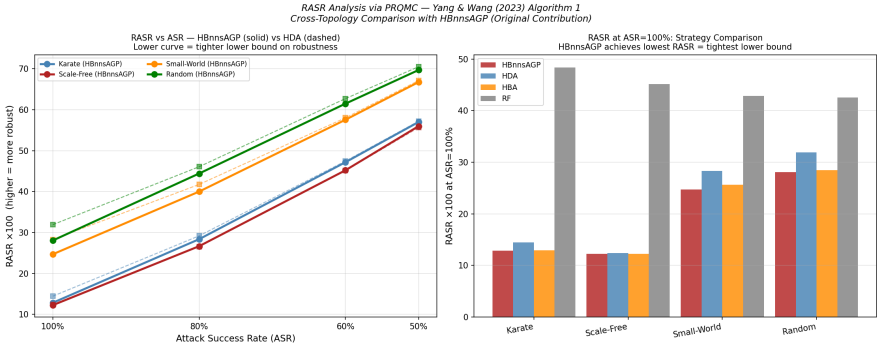


Figure 9: Left: RASR vs. ASR under HBnnsAGP (solid) and HDA (dashed). The four topology lines never cross, confirming the invariant ordering at every ASR level. Right: grouped bar chart at ASR = 100%; HBnnsAGP achieves the lowest RASR in every case.

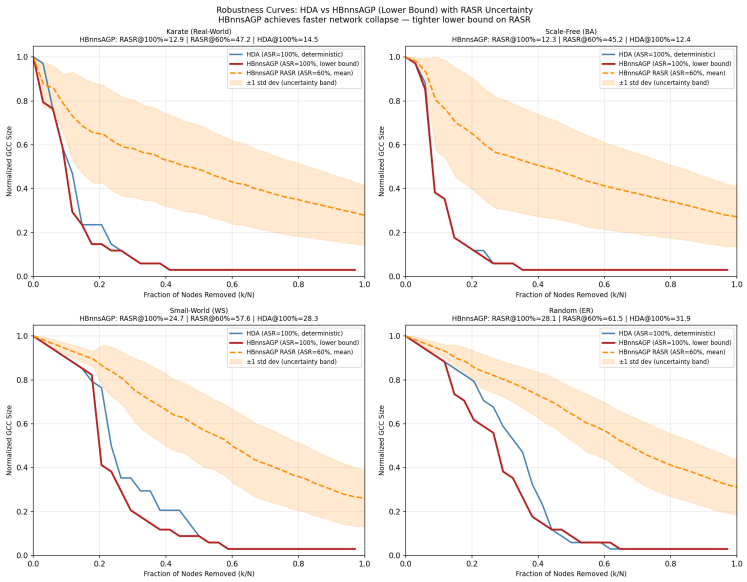


Figure 10: Robustness curves: HDA vs. HBnnsAGP (deterministic, ASR = 100%) and probabilistic RASR (ASR = 60%, $K = 256$). Shaded band: ±1 std. dev. The uncertainty band is widest for scale-free networks at early removal fractions, indicating the most unpredictable collapse behaviour.

same dominant hubs. As ASR decreases below 100%, the advantage diminishes: when attacks frequently fail, the precise sequence matters less than ASR itself [6].

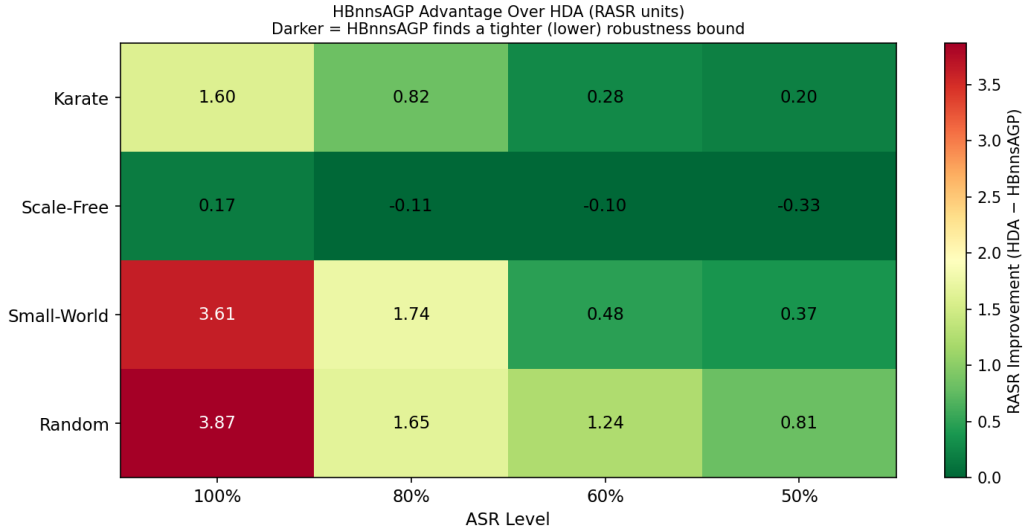


Figure 11: HBnnsAGP advantage over HDA (RASR units: HDA − HBnnsAGP). Darker cells indicate a tighter lower bound. The advantage is greatest for small-world and random networks at high ASR, where bridge nodes are structurally distinct from hubs.

6. Discussion

6.1. Topology as a Fourth Determinant of RASR

The original RASR study [6] showed the metric depends on attack strategy, removal order, and per-node ASR. This analysis adds a fourth factor: the structural class of the network. Table 7 shows the rank ordering is invariant to ASR level, distribution type, and network size. Fig. 1 makes the structural reason clear — the hub-dominated BA network has obvious single points of failure, while the ER network has none.

6.2. Bridge Targeting and the Lower Bound

HBnnsAGP achieves the lowest average RASR (19.47) at full reliability, up to 3.86 units better than HDA on random networks. Bridge nodes found via BCnns are more destructive targets than hubs alone, especially in small-world and random networks where bridges are structurally separate from

hubs. In scale-free networks the two overlap, so the margin shrinks. Below ASR = 100%, HBA can edge ahead on individual networks — when attacks miss often, targeting precision matters less than the failure rate [6].

6.3. Uniform ASR Sensitivity Across Topologies

Dropping ASR from 100% to 50% widens RASR by 42–44 units across all four topologies. A structurally weak network doesn't become safer when attacks fail more often — the absolute level shifts but the ranking doesn't, as Fig. 8 shows.

6.4. Hub-Protection Design Principle

Scale-free networks ($\Delta_{\text{gap}} \approx 0.34$, RASR@100% = 12.27) concentrate fragility in a small hub set, so hub protection gives the highest return. Random networks ($\Delta_{\text{gap}} \approx 0.11$, RASR@100% = 28.07) spread fragility evenly, yielding less benefit. Use the degree heterogeneity ratio $k_{\max}/\langle k \rangle$ as a diagnostic: values $\gg 1$ confirm hub dominance (Fig. 2).

6.5. Connectivity Collapse in Sparse Networks

A WS-equivalent network at $N = 2{,}695$, $\langle k \rangle \approx 2.0$ gives HDA ANC = 0.017 and RF ANC = 0.022 ($\Delta_{\text{gap}} = 0.005$). Near the percolation threshold any removal collapses the network regardless of strategy. Hub protection doesn't help here — only increasing connectivity does.

6.6. Limitations and Future Work

BCnns via exhaustive path enumeration limits HBnnsAGP to $N \leq 200$ here; larger networks need faster path-sampling. RASR is evaluated at $N = 34$ only; extending PRQMC to larger scales needs cluster parallelism. Future work should explore degree-dependent heterogeneous ASR and formal convergence bounds at intermediate ASR levels.

7. Conclusion

Network topology type is an independent, structurally stable determinant of robustness under probabilistic attack. A cross- topology analysis using PRQMC and HBnnsAGP across four network models, five ASR scenarios, and sizes from $N = 34$ to $N = 10{,}670$ yields four main results:

- The fragility ordering Scale-Free < Real-World < Small-World < Random is invariant to ASR level, strategy, and network size — rooted in structural properties visible before any simulation.
- Random networks are 2.29× more robust than scale-free ones, confirmed by both ANC and RASR independently.
- Halving ASR raises RASR by 42–44 units regardless of topology: structural weakness can't be offset by reducing attack reliability.
- Near the percolation threshold, targeted and random attacks become indistinguishable.

Practitioners should compute $k_{\max}/\langle k \rangle$: a heavy- tailed distribution flags a network where protecting a small hub set delivers most of the robustness gain. Future work will extend PRQMC to larger scales and investigate degree-dependent ASR profiles.

Acknowledgment

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.